Equations__inequalities_exponential_and_logarithmic__functions

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Equations_...

GAC 135
(UNIT 2)

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2.0 Equations and Inequalities

2.01 Solving Linear equations Using properties equalities

An **equation** a *is statement that two expressions are equal.*
From the expressions $3(x-1)+x$ and $-x+7$, we can form the equation

$3(x-1)+x=-x+7$

The $(+)$ Property of Equality	**The $(\times)$ Property of Equality**
If x, y, and z represent algebraic expressions and $x=y$ then, $x+z=y+z$	If x, y, and z represent algebraic expressions and $x=y$ then, $xz=yz$

Solving Linear equations using Properties of equality

Example 1

Solve for x: $3(x-1)+x=-x+7$

$3(x-1)+x=-x+7$
$3x-3+x=-x+7$
$4x-3=-x+7$
$5x=10$
$x=2$

$3(x-$

$3(x-1)+x=-x+7$

$\Rightarrow 3x-3+x=-x+7$

$\Rightarrow 3x+x+x=7+3$

$\Rightarrow 5x=10$

$\Rightarrow x=2$

Example 2

Solve for n: $\frac{1}{4}(n+8)-2=\frac{1}{2}(n-6)$

$\frac{1}{4}(n+8)+2=\frac{1}{2}(n-6)$
$\frac{1}{4}n+2-2=\frac{1}{2}n-3$
$4(\frac{1}{4}n)=4(\frac{1}{2}n-3)$
$n=2n-12$
$-n=-12$
$n=12$

2.02 Literal Equations and Formulas

Solving for specified variable in literal equations

A **formula** is an equation that models a known relationship between two or more quantities.
A **literal equation** is simply one that has two or more variables.
Formulas are a type of literal equation, but not every literal equation is a formula.

Example 3: Solving for a specified variable

Given $A=P+PRT$, write P in terms of A, R, and T (solve for P)

$$A=P+PRT$$
$$\Rightarrow A=P(1+RT)$$
$$\Rightarrow \frac{A}{(1+RT)}=\frac{P(1+RT)}{(1+RT)}$$
$$\Rightarrow P=\frac{A}{(1+RT)}$$

$$A = P + PRT$$
$$\Rightarrow A = P(1+RT)$$
$$\Rightarrow \frac{A}{1+RT} = \frac{P(1+RT)}{1+RT}$$
$$\Rightarrow P = \frac{A}{1+RT}$$

Example 4

Given $2x+3y=15$, write y in terms of x (solve for y)

$$2x+3y=15$$
$$3y=15-2x$$
$$y=\frac{(15-2x)}{(3)}$$
$$y=\frac{-2x}{3}x+5$$

$$2x+3y=15$$
$$\Rightarrow 3y=15-2x$$
$$\Rightarrow y=\frac{15-2x}{3}$$

$$y = \frac{(15-2x)}{(3)}$$

$$y = \frac{-2x}{3}x + 5$$

$\rightarrow y = \frac{15-2x}{3}$

$\Rightarrow y = 5 - \frac{2}{3}x.$

Exercise

Solve each equation. Check your answer by substitution

1. $4x + 3(x-2) = 18 - x$
2. $15 - 2x = -4(x+1) + 9$
3. $21 - (2v + 17) = -7 - 3v$
4. $\frac{1}{6}(b+10) - 7 = \frac{1}{3}(b-9)$
5. $\frac{1}{6}(n-12) = \frac{1}{4}(n+8) - 2$

2.10 Linear Inequalities in one variable

Inequalities and solution sets

The set of numbers that satisfy inequality is called **Solution Set**. Instead of using inequality to write solution sets, we will often use

(1) a form of **set notation**,

(2) a **number line** graph or

(3) interval notation

Interval notation is a symbolic way of indicatimg a selected interval of the real number line.
When a number acts as the **boundary point** for an interval(also call **endpoint**), we use left bracket **"["** or the right bracket **"]"** to indicate **inclusion** of end points.
If the boundary point is **not included**, we use a left **"("** or a right parenthesis **")"**

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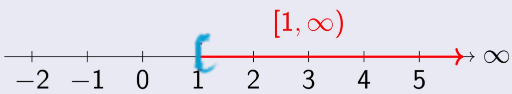
[ ]

Let d represent additional distance: d≥1.

- Set notation : $d|d \geq 1$
- Graph:



- Interval notation: $d \in [1, \infty)$

$d \geq 1$

$[1, \infty)$

$d > 1$ $(1, \infty)$

The (+) Property of Inequality

If x, y and z represent algebraic expressions and $x < y$,

$$x + z < y + z$$

Like quantities(numbers or terms) can be added to both sides of an inequality.

The $(\times)$ Property of inequality

- If x, y, and z represent algebraic expressions and $x < y$:
  - When z is *positive*
    then: $xz < yz$
    (inequality direction remains the same)
  - When z is *negative*
    then: $xz > yz$
    (inequality direction reverses)

$2 < 5$
$2(-3) > (-3)5$
$-6 > -15$

$x < y$
$2 < 5$
$2 - 4 < 5 - 4$
$-2 < 1$
-2 $\quad$ 2

$2 < 5$
$2 \times 3 < 3 \times 5$
$6 < 15$

Example 5: Solving an inequality

Solve the inequality, then graph the solution set and write it in interval notation.
$\frac{-2}{3}x + \frac{1}{2} \leq \frac{5}{6}$

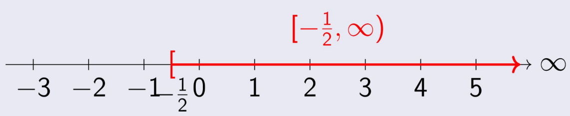
$$\Rightarrow \frac{-2}{3}x + \frac{1}{2} \leq \frac{5}{6}$$
$$\Rightarrow 6(\frac{-2}{3}x + \frac{1}{2}) \leq 6(\frac{5}{6})$$
$$\Rightarrow -4x + 3 \leq 5$$
$$\Rightarrow -4x \leq 2$$
$$-4x \leq 2$$
$$x \geq -\frac{1}{2}$$

$-\frac{2}{3}x + \frac{1}{2} \leq \frac{5}{6}$
$\Rightarrow 6\left[-\frac{2}{3}x + \frac{1}{2}\right] \leq 6(\frac{5}{6})$
$\Rightarrow -4x + 3 \leq 5$
$\Rightarrow -4x \leq 2$
$\Rightarrow -\frac{1}{4}(-4x) \leq -\frac{1}{4}(2)$
$x \geq -\frac{1}{2}$

$-\frac{1}{2}$

$[-\frac{1}{2}, \infty)$

- **Graph:**



- **Interval notation:** $x \in \left[-\frac{1}{2}, \infty\right)$

Exercise 2

Solve the inequality, then graph the solution set and write it in interval notation.

$$\begin{array}{llcl} a. & 7 - (3x + 5) & \geq & 2(x - 4) - 5x \\ b. & 3(x + 4) - 5 & < & 2(x - 3) + x \end{array}$$

Solving Compound Inequalities

In some applications of inequalities, we must consider more than one solution interval. These are called **compound inequalities**, and require us a close look at the operation of **union** " $\bigcup$ " and **intersection** " $\bigcap$ " .

The intersection of two sets X and Y, written $X \bigcap Y$ is the set of all elements *common to both sets*.

The union of two sets X and Y, written $X \bigcup Y$ is the set of all elements *that are in either sets*.

When stating the union of two sets, repetitions are necessary.

Example 6: Finding the Union and Intersection of Two Sets

For a set $X = \{-2, -1, 0, 1, 2, 3\}$ and set $Y = \{1, 2, 3, 4, 5\}$, Determine $X \bigcap Y$ and $X \bigcup Y$

X $\bigcap$ Y is the set of all element in *both X and Y*:
$X \bigcap Y = \{1, 2, 3\}$.

$X \bigcup Y$ is the set of all element in *either X or Y*:
$X \bigcup Y = X = \{-2, -1, 0, 1, 2, 3, 4, 5\}$.

Example 7: Solving a Compound Inequality

Solve the compound inequality, then write the solution in interval notation:
$-3x - 1 < -4$ or $4x + 3 < -6$

First Inequality

$$\begin{aligned} -3x - 1 &< -4 \\ -3x &< -3 \\ x &> 1 \quad \text{(Flipped!)} \end{aligned}$$

Second Inequality

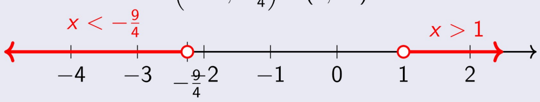
$$\begin{aligned} 4x + 3 &< -6 \\ 4x &< -9 \\ x &< -\frac{9}{4} \end{aligned}$$

Interval notation

$$x \in \left(-\infty, -\frac{9}{4}\right) \cup (1, \infty)$$

Graph

$$\left(-\infty, -\tfrac{9}{4}\right) \cup (1, \infty)$$



- Open circles indicate exclusion of endpoints
- Red lines show solution intervals
- Disjoint intervals require "or" condition

Applications of Inequalities

Domain and Allowable Values

The set of allowable values is referred to as the **domain** of the expression.

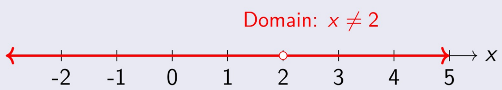
The allowable values are said to be *"in the domain"* of the expression; values that are not allowed is said to be *"outside the domain"*

When the denominator of a fraction contains a variable expression, values that cause a denominator of zero are outside the domain.

Example 8: Determining the domain of an Expression

Determine the domain of the expression $\dfrac{6}{x-2}$. State the result in set notation, graphically and using interval notation.

- Denominator must not be zero: $x - 2 \neq 0 \Rightarrow x \neq 2$
- **Set notation:** $\{x \in \mathbb{R} \mid x \neq 2\}$
- **Interval notation:** $(-\infty, 2) \cup (2, \infty)$

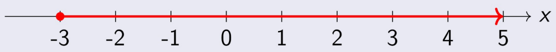


Example 9: Determining the domain of an Expression

Determine the domain of the expression $\sqrt{x+3}$. State the result in set notation, graphically and using interval notation.

- Expression under square root must be non-negative: $x + 3 \geq 0 \Rightarrow x \geq -3$
- **Set notation:** $\{x \in \mathbb{R} \mid x \geq -3\}$
- **Interval notation:** $[-3, \infty)$

Domain: $x \geq -3$



EXERCISE 2

Graph Each Inequality on a Number Line

1. $x \geq -2$
2. $x = -3$
3. $m \leq 5$
4. $x \neq 1$
5. $x \leq 7$
6. $x \neq -3$
7. $5 < x$
8. $-3 < y \leq 4$

Solve Inequality and Graph the Solution

9. $5a - 11 \geq 2a - 5$
10. $-8n + 5 > -2n - 12$
11. $2(n + 3) - 4 > 5n$
12. $5(x + 2) - 3 < 5x + 11$
13. $\frac{x}{8} + 4 < -4$
14. $\frac{2y}{5} + 10 < -2$

Solve and Write in Set Notation

15. $7 - 2(x + 3) = 4x - 6(x - 3)$
16. $5(k - 6) - 5 = 24k - 3k + 1$
17. $4(3x - 9) + 18 \leq 2(5x + 1) + 2x$
18. $8 - 6(1 - 5m) = -9m - (3 - 4m)$
19. $-6(p - 1) + 2p \leq 2(2p - 3)$
20. $9(w - 1) - 3w = 2(5 - 3w) + 1$

Intersection and Union of Sets

Given:

$$A = \{-1, 0, 2, 3\}, \quad B = \{-1, 1, 2\},$$
$$C = \{-2, 0, 1, 2\}, \quad D = \{1, 3, 4\}$$

21. $A \cap B$, $A \cup B$
22. $A \cap C$, $A \cup C$
23. $B \cap C$, $B \cup C$
24. $C \cap D$, $C \cup D$
25. $B \cap D$, $B \cup D$

Compound Inequalities: Graph Interval

26. $x < -2$ or $x > 1$
27. $x \leq -5$ or $x > 3$
28. $x < 5$ and $x > 1$
29. $x \geq -4$ and $x < 3$
30. $x = 5$ and $x \leq 1$
31. $x = 5$ and $x \leq -7$

Solve Compound Inequalities and Graph

32. $4(x-1) \leq 20$ or $x+6 > 9$
33. $-3(x+2) > 15$ or $x-3 \leq -1$
34. $-2n+7 \leq 3$ and $2x \leq 0$
35. $-3x+5 \leq 17$ and $5x \leq 0$
36. $2g \leq 8$ and $-4x > -1$
37. $g+5 \leq 0$ and $-3x < -2$

More Compound Inequalities

38. $\frac{3x}{4} \leq -3$ or $-3x+1 > 5$
39. $\frac{5}{10} \leq -2x-3$ or $3 > 2$
40. $-2x+5 < 7$ and $x \leq 7$
41. $2 \leq 3x-4 \leq 19$
42. $-0.5 \leq 0.3x \leq 1.7$

2.20 Absolute Value Equation and Inequalities

While the equations $x+1=5$ and $|x+1|=5$ are similar in many respects, note the first has only the solution $x=4$, while either $x=4$ or $x=-6$ will satisfy the second.

The fact there are two solutions shouldn't surprise us, as it's a natural result of how absolute value is defined.

Solving Absolute Value Equations

The absolute value of a number x can be thought of as its distance from zero on the number line, regardless of direction. This means:

$$|x| = 4 \quad \Rightarrow \quad x = -4 \quad \text{or} \quad x = 4$$

Property of Absolute Value Equations

If X represents an algebraic expression and k is a positive real number, then:

$$|X| = k \quad \Longleftrightarrow \quad X = -k \quad \text{or} \quad X = k$$

Worthy of Note

- If $k < 0$, the equation $|X| = k$ has **no solution** since absolute value is never negative.
- If $k = 0$, then $|X| = 0$ has **only one solution**: $X = 0$.

Example 6

Solve: $\left|5 - \frac{2}{3}x\right| - 9 = 8$

$$\left|5 - \frac{2}{3}x\right| = 17 \qquad \text{Isolate absolute value}$$

$$5 - \frac{2}{3}x = 17 \quad \text{or} \quad 5 - \frac{2}{3}x = -17 \qquad \text{Apply property}$$

$$-\frac{2}{3}x = 12 \quad \text{or} \quad -\frac{2}{3}x = -22$$

$$x = -18 \quad \text{or} \quad x = 33 \qquad \text{Multiply by } -\frac{3}{2}$$

Checking solutions

For $x = 33$

$$\begin{aligned}\left|5 - \frac{2}{3}(33)\right| - 9 &= 8\\ |5 - 22| - 9 &= 8\\ |-17| - 9 &= 8\\ 17 - 9 &= 8\\ 8 &= 8 \quad \checkmark\end{aligned}$$

For $x = -18$

$$\begin{aligned}\left|5 - \frac{2}{3}(-18)\right| - 9 &= 8\\ |5 + 12| - 9 &= 8\\ |17| - 9 &= 8\\ 17 - 9 &= 8\\ 8 &= 8 \quad \checkmark\end{aligned}$$

Solution Set

Both solutions are valid: $\{-18, 33\}$

Multiplicative Property

Property

For any algebraic expressions A and B:

$$|AB| = |A||B|$$

Example 7

Solve: $|-2x| + 5 = 13$

$$\begin{aligned}|-2x| &= 8\\ |-2|\,|x| &= 8 \qquad \text{Apply multiplicative property}\\ 2|x| &= 8\\ |x| &= 4\\ x &= 4 \quad \text{or} \quad x = -4\end{aligned}$$

Worthy of Note

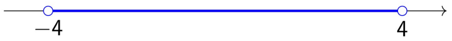
- Always isolate the absolute value first before applying properties
- The same principle applies to absolute value inequalities
- Checking solutions is crucial as some may be extraneous

Solving "Less Than" Absolute Value Inequalities

Absolute value inequalities can be solved using the basic concept underlying the property of absolute value equalities.

- $|x| = 4$ asks for all numbers x whose distance from zero is equal to 4
- $|x| < 4$ asks for all numbers x whose distance from zero is less than 4

$$-4 < x < 4$$



Property I: Absolute Value Inequalities

If X represents an algebraic expression and k is a positive real number:

$$|X| < k \quad \text{implies} \quad -k < X < k$$

Example 9

Part (a)

Solve: $\dfrac{|3x+2|}{4} \leq 1$

$\lvert 3x+2\rvert \leq 4$	Multiply both sides by 4
$-4 \leq 3x+2 \leq 4$	Apply Property I
$-6 \leq 3x \leq 2$	Subtract 2 from all parts
$-2 \leq x \leq \dfrac{2}{3}$	Divide by 3

Worthy of Note

Test $x = 0$ from the solution interval:

$$\frac{|3(0)+2|}{4} = \frac{1}{2} \leq 1 \quad \checkmark$$

Part (b)

Solve: $|2x - 7| < -5$

- Absolute value expressions are always non-negative: $|X| \geq 0$
- No real number satisfies $|X| <$ negative
- Solution: No solution ($\emptyset$)

Upcoming: "Greater Than" Inequalities

For $|x| > 4$, solutions are:

- In the interval left of -4 **or**
- In the interval right of 4

These form disjoint intervals.

Key Concepts Summary

For $|X| < k$ where $k > 0$

- Rewrite as $-k < X < k$
- Graphically: single interval between $-k$ and k
- Endpoints not included (strict inequality)

Special Cases

- If $k \leq 0$ in $|X| < k$:
  - No solution when $k < 0$
  - Only $X = 0$ when $k = 0$

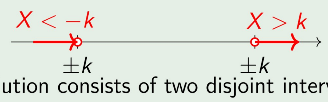
Property II: Absolute Value Inequalities

If X represents an algebraic expression and k is a positive real number,

$$|X| > k \quad \text{implies} \quad X < -k \ \textbf{or} \ X > k$$

Example

Graphical Representation



Solution consists of two disjoint intervals

Key Features

- Used for "greater than" inequalities
- Produces **two separate solution intervals**
- Inequality sign points away from zero in both directions

Exercise 3

Concepts and Vocabulary

Fill in the blank with the appropriate word or phrase. Carefully reread the section if needed.

1. When multiplying or dividing by a negative quantity, we __________ the inequality to maintain a true statement.
2. To write an absolute value equation or inequality in simplified form, we __________ the absolute value expression on one side.
3. The absolute value equation $|2x + 3| = 7$ is true when $2x + 3 =$ __________ or when $2x + 3 =$ __________.
4. The absolute value inequality $|3x - 6| < 12$ is true when $3x - 6 >$ __________ and $3x - 6 <$ __________.

Describe each solution set (assume $k > 0$). Justify your answer.

5. $|ax + b| < -k$
6. $|ax + b| > -k$

Solve each absolute value equation. Write the solution in set notation.

7. $2|m - 1| - 7 = 3$
8. $3|n - 5| - 14 = -2$
9. $-3|x + 5| + 6 = -15$
10. $-2|y + 3| - 4 = -14$
11. $2|4v + 5| - 6.5 = 10.3$
12. $7|2w + 5| + 6.3 = 11.2$
13. $-|7p - 3| + 6 = -5$

Solve each absolute value inequality. Write solutions in interval notation.

14. $|x - 2| \leq 7$
15. $|y + 1| \leq 3$
16. $-3|m| - 2 > 4$
17. $-2|n| + 3 > 7$
18. $\frac{|5v+1|}{4} + 8 < 9$
19. $\frac{|3w-2|}{2} + 6 < 8$
20. $3|p + 4| + 5 < 8$
21. $5|q - 2| - 7 \leq 8$
22. $|3b - 11| + 6 \leq 9$
23. $|2c + 3| - 5 < 1$
24. $|4 - 3z| + 12 < 7$
25. $|2 - 7u| + 7 \leq 4$

Exponential Functions

In general, exponential functions are defined as follows.

For $b > 0$, $b \neq 1$, and all real numbers x,

$$f(x) = b^x$$

defines the **base b exponential function**.

Limiting b to positive values ensures that outputs will be real numbers, and the restriction $b \neq 1$ is needed since $y = 1^x$ is a constant function (1 raised to any power is still 1). Specifically note the domain of an exponential function is all real numbers, and that all of the familiar properties of exponents still hold. A summary of these properties follows.

Exponential Properties

For real numbers a, b, m, and n, with $a, b > 0$,

$$b^m \cdot b^n = b^{m+n} \qquad \frac{b^m}{b^n} = b^{m-n}$$

$$(b^m)^n = b^{mn} \qquad (ab)^n = a^n \cdot b^n$$

$$b^{-n} = \frac{1}{b^n} \qquad \left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

EXAMPLE 11 Evaluating Exponential Functions

Evaluate each exponential function for $x = 2$, $x = -1$, $x = \frac{1}{2}$, and $x = \pi$. Use a calculator for $x = \pi$, rounding to five decimal places.

a. $f(x) = 4^x$

$$f(2) = 4^2 = 16$$

$$f(-1) = 4^{-1} = \frac{1}{4}$$

$$f\left(\frac{1}{2}\right) = 4^{\frac{1}{2}} = \sqrt{4} = 2$$

$$f(\pi) = 4^\pi \approx 77.88023$$

b. $g(x) = \left(\frac{4}{9}\right)^x$

Properties of Logarithms; Solving Exponential Logarithmic Equations

Learning Objectives

In Section you will learn how to:

A. Solve logarithmic equations using the fundamental properties of logarithms
B. Apply the product, quotient, and power properties of logarithms
C. Solve general logarithmic and exponential equations
D. Solve applications involving logistic, exponential, and logarithmic functions

Solving Equations Using the Fundamental Properties of Logarithms

Conversion expressions from exponential form to logarithmic form using the basic definition:

$$x = b^y \Leftrightarrow y = \log_b x$$

This relationship reveals the following four properties:

Fundamental Properties of Logarithms

For any base $b > 0, b \neq 1$,

I. $\log_b b = 1$, since $b^1 = b$
II. $\log_b 1 = 0$, since $b^0 = 1$
III. $\log_b b^x = x$, since $b^x = b^x$ (exponential form)
IV. $b^{\log_b x} = x$, since $\log_b x = \log_b x$ (logarithmic form)

Inverse Relationship

Also note that Properties III and IV demonstrate that $y = \log_b x$ and $y = b^x$ are inverse functions. In common language, "a base-b logarithm undoes a base-b exponential," and "a base-b exponential undoes a base-b logarithm." For $f(x) = \log_b x$ and $f^{-1}(x) = b^x$, using composition verifies the inverse relationship:

$$\begin{array}{ll} (f \circ f^{-1})(x) = f[f^{-1}(x)] & (f^{-1} \circ f)(x) = f^{-1}[f(x)] \\ = \log_b b^x & = b^{\log_b x} \\ = x & = x \end{array}$$

These properties can be used to solve basic equations involving logarithms and exponentials.

Example 10:Solving Basic Logarithmic Equations

Solve each equation by applying fundamental properties. Answer in exact form and approximate form using a calculator (round to 1000ths).

a. $\ln x = 2$

$$\begin{aligned} \ln x &= 2 && \text{given} \\ e^{\ln x} &= e^2 && \text{exponentiate both sides} \\ x &= e^2 && \text{Property IV} \\ &\approx 7.389 && \text{result} \end{aligned}$$

b. $-0.52 = \log x$

$$\begin{aligned} -0.52 &= \log x && \text{given} \\ 10^{-0.52} &= 10^{\log x} && \text{exponentiate both sides} \\ x &= 10^{-0.52} && \text{Property IV} \\ &\approx 0.302 && \text{result} \end{aligned}$$

Fundamental Properties of Logarithms

For positive real numbers X, Y, b $(b \neq 1)$ and any real exponent p:

Product Property

$\log_b(XY) = \log_b X + \log_b Y$

The logarithm of a product equals the sum of the logarithms

Quotient Property

$\log_b\left(\frac{X}{Y}\right) = \log_b X - \log_b Y$

The logarithm of a quotient equals the difference of the logarithms

Power Property

$\log_b X^p = p \log_b X$

The logarithm of a power equals the exponent times the logarithm

ransform products to sums, quotients to differences, and powers to products

Change-of-Base Formula

For positive real numbers X, a, b $(a, b \neq 1)$

$\log_b X = \dfrac{\log X}{\log b}$	$\log_b X = \dfrac{\ln X}{\ln b}$	$\log_b X = \dfrac{\log_a X}{\log_a b}$
base 10	base e	arbitrary base a

Proof of the Change-of-Base Formula

$$\begin{aligned}
\text{Let } y &= \log_b X \\
b^y &= X && \text{(Convert to exponential form)} \\
\log_a(b^y) &= \log_a X && \text{(Take base-}a\text{ log)} \\
y \log_a b &= \log_a X && \text{(Power property)}
\end{aligned}$$

$$\begin{aligned}
y &= \frac{\log_a X}{\log_a b} && \text{(Solve for } y\text{)} \\
\log_b X &= \frac{\log_a X}{\log_a b} && \text{(Substitute back)}
\end{aligned}$$

Uniqueness Property of Logarithms

If the simplified form of an equation yields a logarithmic term on both sides, the uniqueness property of logarithms provides an efficient way to work toward a solution. Since logarithmic functions are one-to-one, we have:

The Uniqueness Property of Logarithms

For positive real numbers m, n, and $b \neq 1$:

$$\text{If } \log_b m = \log_b n, \quad \text{then } m = n$$

Equal bases imply equal arguments.

EXERCISE 4

1. Solve Each Equation by Applying Fundamental Properties, (Round to thousandths)

Basic Logarithmic/Exponential Equations

7. $\ln x = 3.4$
8. $\ln x = \frac{1}{2}$
9. $\log x = \frac{1}{4}$
10. $\log x = 1.6$
11. $e^x = 9.025$
12. $e^y = 0.343$
13. $10^x = 18.197$
14. $10^x = 0.024$

More Complex Equations

15. $4e^{x-2} + 5 = 70$
16. $2 - 3e^{0.4x} = -7$
17. $10^x + 5 - 228 = -150$
18. $10^x + 27 = 190$
19. $-150 = 290.8 - 190e^{-0.75x}$
20. $250e^{0.05x+1} + 175 = 1175$

2. Solve Each Equation

Write answers in exact form and approximate form to four decimal places

Logarithmic Equations

21. $3\ln(x+4) - 5 = 3$
22. $-15 = -8\ln(3x) + 7$
23. $-1.5 = 2\log(5-x) - 4$
24. $-4\log(2x) + 9 = 3.6$
25. $\frac{1}{2}\ln(2x+5) + 3 = 3.2$
26. $\frac{3}{4}\ln(4x) - 6.9 = -5.1$

3. Properties of Logarithms

Write each expression as a single term

Sum of Logarithms

27. $\ln(2x) + \ln(x-7)$
28. $\ln(x+2) + \ln(3x)$
29. $\log(x+1) + \log(x-1)$
30. $\log(x-3) + \log(x+3)$
39. $\log_2 7 + \log_2 6$
40. $\log_2 2 + \log_2 15$

Difference of Logarithms

33. $\log x - \log(x+1)$
34. $\log(x-2) - \log x$
35. $\ln(x-5) - \ln x$
36. $\ln(x+3) - \ln(x-1)$
37. $\ln(x^2-4) \pm \ln(x+2)$
38. $\ln(x^2-25) - \ln(x+5)$

4. Power Property Applications

Rewrite each term

Using the Power Property

43. $\log 8^{x+2}$
44. $\log 15^{x-3}$
45. $\ln 5^{2x-1}$
46. $\ln 10^{3x+2}$
47. $\log \sqrt{22}$
48. $\log \sqrt[3]{34}$
49. $\log_3 81$
50. $\log_7 121$

Special Cases

41. $\log_3(x^2 - 2x) + \log_3 x^{-1}$
42. $\log_3(3x^2 + 5x) - \log_3 x$

Use properties of logarithms to write as sum/difference of simple logarithmic terms

51. $\log(a^3 b)$
52. $\log(m^2 n)$
53. $\ln(\sqrt[3]{y})$
54. $\ln(\sqrt[3]{pq})$
55. $\ln\left(\frac{x^2}{y}\right)$
56. $\ln\left(\frac{m^2}{n^3}\right)$
57. $\log\left(\sqrt{\frac{x-2}{x}}\right)$
58. $\log\left(\sqrt[3]{\frac{3-y}{2y}}\right)$

Solve each equation using uniqueness property of logarithms

81. $\log(5x+2) = \log 2$
82. $\log(2x-3) = \log 3$
83. $\log_4(x+2) - \log_4 3 = \log_4(x-1)$
84. $\log_3(x+6) - \log_3 x = \log$
85. $\ln(8x-4) = \ln 2 + \ln x$
86. $\ln(x-1) + \ln 6 = \ln(3x)$

Uniqueness Property

For positive real numbers M, N, b $(b \neq 1)$:

$$\text{If } \log_b M = \log_b N, \text{ then } M = N$$